



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

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Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Experiment No.: 02

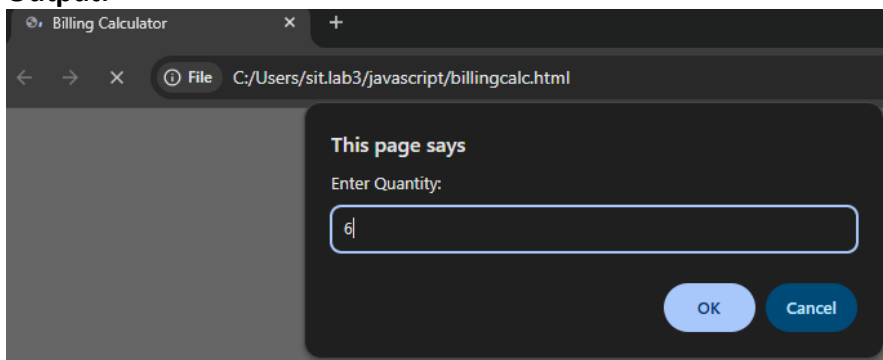
1. **Title:** Use var, let, const, and demonstrate basic features like template literals and destructuring, build a billing calculator with user input.

2. **Software/Tools Required:** HTML, VS Code.

3. **Theory:** JavaScript is a scripting language used to make web pages interactive and dynamic. It allows developers to perform calculations, validate user input, display messages, and manipulate webpage content. In this experiment, a **Billing Calculator** is developed using JavaScript. The program accepts user inputs such as customer name, product name, quantity, and price per item through dialog boxes (prompt()). It then calculates the total bill and displays a formatted billing receipt on the webpage. The program demonstrates the following JavaScript features:

- **var** – Used to declare function-scoped variables. It can be redeclared and updated.
- **let** – Used to declare block-scoped variables. It can be updated but not redeclared within the same scope.
- **const** – Used to declare constants whose values cannot be reassigned.
- **Template Literals** – Enclosed within backticks (`), they allow embedding variables directly into strings using `\${}` for easier string formatting.
- **Destructuring** – A feature that extracts values from arrays or objects into separate variables, making the code cleaner and more readable.

4. **Output:**





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← ↻ 127.0.0.1:5500/index.html/mahimadugar

Billing Calculator

Billing Receipt

Customer: mahima

Product: laptop

Quantity: 3

Price per Item: Rs40000

Total: Rs120000

GST (18%): RS.21600.00

Final Bill: Rs.141600.00



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5. Case Study Title: Develop a webpage for an online shopping website using HTML, CSS, and HTML5 tags, and calculate the bill.

6. Case Study Program Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Online Shopping Bill Calculator</title>

<style>
body{
    font-family: Arial, sans-serif;
    background:#f2f2f2;
}

.container{
    width:500px;
    margin:30px auto;
    background:white;
    padding:20px;
    border-radius:10px;
    box-shadow:0 0 10px gray;
}

h2{
    text-align:center;
    color:darkblue;
}

label{
    display:block;
    margin-top:10px;
    font-weight:bold;
}

input,select{
    width:100%;
    padding:8px;
    margin-top:5px;
}
```



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```
button{
  width:100%;
  padding:10px;
  margin-top:20px;
  background:green;
  color:white;
  border:none;
  font-size:16px;
  cursor:pointer;
}
```

```
button:hover{
  background:darkgreen;
}
```

```
#result{
  margin-top:20px;
  padding:15px;
  background:#e8f5e9;
  border-radius:5px;
}
```

```
</style>
</head>
```

```
<body>
```

```
<div class="container">
```

```
<h2>Online Shopping Bill Calculator</h2>
```

```
<label>Customer Name</label>
```

```
<input type="text" id="name">
```

```
<label>Mobile Number</label>
```

```
<input type="text" id="mobile">
```

```
<label>Invoice Number</label>
```

```
<input type="text" id="invoice">
```

```
<label>Product Name</label>
```

```
<input type="text" id="product">
```

```
<label>Quantity (Kg)</label>
```

```
<input type="number" id="qty">
```

```
<label>Rate per Kg (₹)</label>
```

```
<input type="number" id="rate">
```



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```
<label>Discount (%)</label>
<input type="number" id="discount">

<label>GST (%)</label>
<input type="number" id="gst">

<label>Packing Charges (₹)</label>
<input type="number" id="packing">

<label>Payment Mode</label>
<select id="payment">
  <option>Cash</option>
  <option>Card</option>
  <option>UPI</option>
</select>

<label>Membership</label>
<select id="member">
  <option>Yes</option>
  <option>No</option>
</select>

<button onclick="calculateBill()">Generate Bill</button>

<div id="result"></div>

</div>

<script>

function calculateBill(){

  // String Data Types
  var customerName = document.getElementById("name").value;
  var mobile = document.getElementById("mobile").value;
  var invoice = document.getElementById("invoice").value;
  var product = document.getElementById("product").value;

  // Number Data Types
  let qty = Number(document.getElementById("qty").value);
  let rate = Number(document.getElementById("rate").value);
  let discount = Number(document.getElementById("discount").value);
  let gst = Number(document.getElementById("gst").value);
  let packing = Number(document.getElementById("packing").value);

  // Constant
  const payment = document.getElementById("payment").value;
```



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```
// Boolean Data Type
let membership = document.getElementById("member").value=="Yes";
```

```
// Undefined and Null
let remarks;
let coupon = null;
```

```
let amount = qty * rate;
```

```
let discountAmount = amount * discount / 100;
```

```
let afterDiscount = amount - discountAmount;
```

```
let gstAmount = afterDiscount * gst / 100;
```

```
let total = afterDiscount + gstAmount + packing;
```

```
if(membership){
    total = total - (total * 5 / 100);
}
```

```
document.getElementById("result").innerHTML = `
<h3>Invoice</h3>
```

```
Customer Name : ${customerName}<br>
Mobile Number : ${mobile}<br>
Invoice Number : ${invoice}<br>
Product Name : ${product}<br><br>
```

```
Quantity : ${qty} Kg<br>
Rate : ₹${rate}<br>
Amount : ₹${amount.toFixed(2)}<br>
Discount : ₹${discountAmount.toFixed(2)}<br>
GST : ₹${gstAmount.toFixed(2)}<br>
Packing Charges : ₹${packing}<br>
Membership : ${membership}<br>
Payment Mode : ${payment}<br><br>
```

```
<h3>Total Bill : ₹${total.toFixed(2)}</h3>
```

```
<hr>
```

```
<b>Primitive Data Types Used</b><br>
String : ${typeof customerName}<br>
Number : ${typeof qty}<br>
Boolean : ${typeof membership}<br>
Undefined : ${typeof remarks}<br>
Null : ${coupon}
```



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```
;
}

</script>

</body>
</html>
```

7. Output:

← → ↻ C:/Users/Hp/Javascript/billing.html/mahimadugar

Online Shopping Bill Calculator

Customer Name	Mahima Dugar
Mobile Number	8208239478
Invoice Number	123
Product Name	tea leaves
Quantity (Kg)	3
Rate per Kg (₹)	150
Discount (%)	5
GST (%)	2.5
Packing Charges (₹)	12
Payment Mode	Cash
Membership	Yes

Generate Bill



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Generate Bill

Invoice
Customer Name : Mahima Dugar
Mobile Number : 8208239478
Invoice Number : 123
Product Name : tea leaves

Quantity : 3 Kg
Rate : ₹150
Amount : ₹450.00
Discount : ₹22.50
GST : ₹10.69
Packing Charges : ₹12
Membership : true
Payment Mode : Cash

Total Bill : ₹427.68

Primitive Data Types Used
String : string
Number : number
Boolean : boolean
Undefined : undefined
Null : null

- 5. Result/Conclusion:** The Billing Calculator was successfully implemented using JavaScript. The program accepts user input, performs the billing calculation, and displays the receipt correctly. It successfully demonstrates the use of **var**, **let**, **const**, **template literals**, and **destructuring** while providing a simple and interactive billing application.



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Experiment No.: 03

- Title:** Implement control structures and form validation; create a grading system based on user-entered marks.
- Software/Tools Required:** HTML, VS Code.
- Theory:** This experiment demonstrates the use of control structures (if-else statements) and form validation in JavaScript. The user enters marks through an HTML form, and the program checks whether the input is valid (between 0 and 100). Based on the entered marks, the program determines the student's grade using conditional statements and displays the result dynamically on the webpage. This experiment helps in understanding decision-making, input validation, and user interaction in web development.
- Output:**

file:///C:/Users/sit.lab3/Desktop/jslab/webpage/prac3.html/mahimadugar

Mahima's Student Grading System (24070521280):

Enter Marks:

Marks = 90

Grade = A



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5. Case Study Title:

Task 1: Scholarship Eligibility

Case Study:

A college awards scholarships based on marks.

Below 50 - Not Eligible

50–69 - Bronze

70–84 - Silver

85–94 - Gold

95–100 - Platinum

Task:

Write a JavaScript program that accepts student marks and displays the scholarship category.

6. Case Study Program Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Scholarship Eligibility</title>

  <style>
    body {
      font-family: Arial, sans-serif;
      background: #f2f2f2;
      display: flex;
      justify-content: center;
      align-items: center;
      height: 100vh;
    }

    .container {
      background: white;
      padding: 25px;
      border-radius: 10px;
      box-shadow: 0 0 10px rgba(0,0,0,0.2);
      width: 350px;
      text-align: center;
    }

    h2 {
      color: #2c3e50;
    }
  </style>
</head>
<body>
  <div class="container">
    <h2>Scholarship Eligibility</h2>
    <div class="form">
      <input type="text" value="Enter Marks" />
      <input type="button" value="Check Eligibility" />
    </div>
    <div class="result">
      <div class="text">Result</div>
      <div class="text">Category</div>
    </div>
  </div>
</body>
</html>
```



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```
input {
  width: 90%;
  padding: 10px;
  margin: 10px 0;
  font-size: 16px;
}

button {
  padding: 10px 20px;
  background: green;
  color: white;
  border: none;
  cursor: pointer;
  font-size: 16px;
  border-radius: 5px;
}

button:hover {
  background: darkgreen;
}

#result {
  margin-top: 20px;
  font-size: 18px;
  font-weight: bold;
  color: blue;
}
</style>
</head>
<body>

<div class="container">
  <h2>Scholarship Eligibility</h2>

  <label>Enter Student Marks:</label><br>
  <input type="number" id="marks" placeholder="Enter marks (0-100)">

  <br>
  <button onclick="checkScholarship()">Check</button>

  <p id="result"></p>
</div>

<script>
function checkScholarship() {

  let marks = Number(document.getElementById("marks").value);
```



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```
let scholarship = "";

if (marks < 0 || marks > 100) {

scholarship = "Please enter valid marks (0-100).";

}
else if (marks < 50) {
    scholarship = "Not Eligible";
}
else if (marks <= 69) {
    scholarship = "Bronze Scholarship";
}
else if (marks <= 84) {
    scholarship = "Silver Scholarship";
}
else if (marks <= 94) {
    scholarship = "Gold Scholarship";
}
else {
    scholarship = "Platinum Scholarship";
}

document.getElementById("result").innerHTML =
    "Scholarship Category: " + scholarship;
}
</script>

</body>
</html>
```

7. Output:

127.0.0.1:5500/calci/grading.html/mahimadugar

Scholarship Eligibility

Enter Student Marks:

78

Check

**Scholarship Category: Silver
Scholarship**



॥ वसुधैव कुटुम्बकम् ॥

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8. Result/Conclusion: The grading system successfully accepted the user's marks, validated the input, and displayed the correct grade according to the predefined grading criteria. This experiment helped in understanding how form validation ensures valid user input and how control structures (if-else statements) are used to make decisions and generate appropriate output in a web application.



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Experiment No.: 04

- 1. Title:** Implement control structures and form validation; create a grading system based on user-entered marks.
- 2. Software/Tools Required:** HTML, VS Code.
- 3. Theory:** JavaScript provides different types of functions, such as function declarations, function expressions, and arrow functions, to perform reusable tasks efficiently. Variable scope controls where variables can be accessed and includes global, function, and block scope. A closure is a feature in which an inner function remembers and accesses variables from its outer function even after the outer function has finished execution. JavaScript also supports exception handling through the try-catch statement, which helps detect and manage runtime errors without stopping the program. In this practical, these concepts are applied to develop a palindrome checker, which verifies whether a given string or number reads the same forwards and backwards, demonstrating the practical use of functions, scope, closures, and error handling.

4. Output:

/mahimadugar

Palindrome Checker

Check

"level" is a Palindrome ✓
Checks Performed: 2



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CASE STUDY-04

Practical 4: ATM Card PIN Verification Using JavaScript Functions

1. Aim:

To verify whether an ATM PIN is a palindrome using JavaScript functions and demonstrate function declaration, function expression, arrow function, and scope.

2. Case Study Program Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>ATM PIN Verification</title>
```

```
<style>
*{
  margin:0;
  padding:0;
  box-sizing:border-box;
  font-family:Arial, Helvetica, sans-serif;
}

body{
  height:100vh;
  display:flex;
  justify-content:center;
  align-items:center;
  background:linear-gradient(135deg,#0f2027,#203a43,#2c5364);
}

.container{
  width:400px;
  background:white;
  padding:30px;
  border-radius:20px;
  text-align:center;
  box-shadow:0 10px 30px rgba(0,0,0,0.4);
}
```



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```
.icon{
  font-size:60px;
}

h2{
  color:#0d47a1;
  margin:10px 0;
}

p{
  color:gray;
  margin-bottom:20px;
}

input{
  width:100%;
  padding:12px;
  border:2px solid #2196F3;
  border-radius:10px;
  font-size:18px;
  text-align:center;
  outline:none;
}

input:focus{
  border-color:#4CAF50;
}

button{
  width:100%;
  margin-top:20px;
  padding:12px;
  border:none;
  border-radius:10px;
  background:#1565c0;
  color:white;
  font-size:18px;
  cursor:pointer;
  transition:.3s;
```



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}

```
button:hover{
  background:#0d47a1;
  transform:scale(1.05);
}
```

```
#output{
  margin-top:25px;
  padding:15px;
  border-radius:10px;
  display:none;
}
```

```
.success{
  background:#d4edda;
  color:#155724;
  border:2px solid #28a745;
}
```

```
.error{
  background:#f8d7da;
  color:#721c24;
  border:2px solid #dc3545;
}
```

```
.footer{
  margin-top:20px;
  color:#666;
  font-size:14px;
}
```

</style>

</head>

<body>

<div class="container">

<div class="icon"> </div>



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<h2>ATM PIN Verification</h2>

<p>Enter your 4-digit ATM PIN</p>

<input type="password" id="pin" placeholder="Enter PIN">

<button onclick="checkPIN()">Verify PIN</button>

<div id="output"></div>

<div class="footer">
ABC Bank Security System
</div>

</div>

<script>

// Global Variable
let bankName="ABC Bank";

// Function Declaration
function reversePIN(pin){
 return pin.split("").reverse().join("");
}

// Function Expression
const isPalindrome=function(pin){
 return pin===reversePIN(pin);
}

// Arrow Function
const displayResult=(pin)=>{

let output=document.getElementById("output");

if(isPalindrome(pin)){
 output.className="success";
 output.innerHTML=



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```
"<h3>■ PIN Verified</h3>" +
"<p><b>Bank:</b> "+bankName+"</p>" +
"<p><b>PIN:</b> "+pin+"</p>" +
"<p>🇮🇳 Your PIN is a <b>Palindrome</b>.</p>";
}
else{
output.className="error";
output.innerHTML=
"<h3>+ Invalid Pattern</h3>" +
"<p><b>Bank:</b> "+bankName+"</p>" +
"<p><b>PIN:</b> "+pin+"</p>" +
"<p>Your PIN is <b>Not a Palindrome.</b></p>";
}

output.style.display="block";

}

function checkPIN(){

let pin=document.getElementById("pin").value;

if(pin==""){
alert("Please enter your PIN.");
return;
}

displayResult(pin);

}

</script>

</body>
</html>
```



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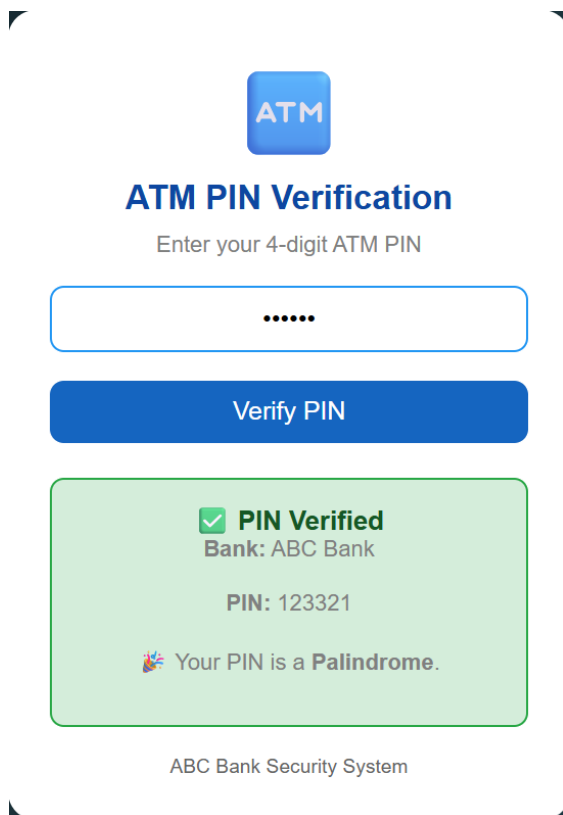
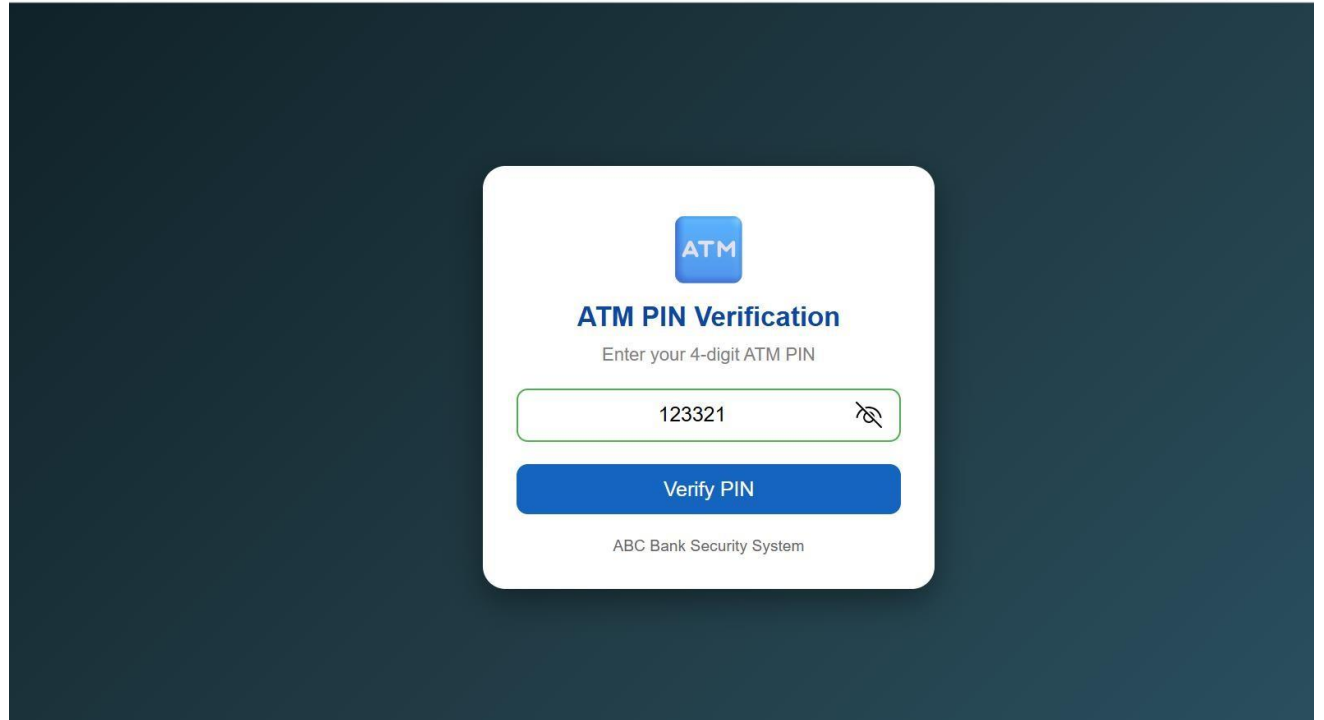
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3. Output:

127.0.0.1:5500/ATM/palin.html/mahimadugar





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5.Result: The program was executed successfully. It demonstrated different types of JavaScript functions, variable scope, closures, and exception handling using try-catch. The palindrome checker correctly identified whether the given input was a palindrome or not.



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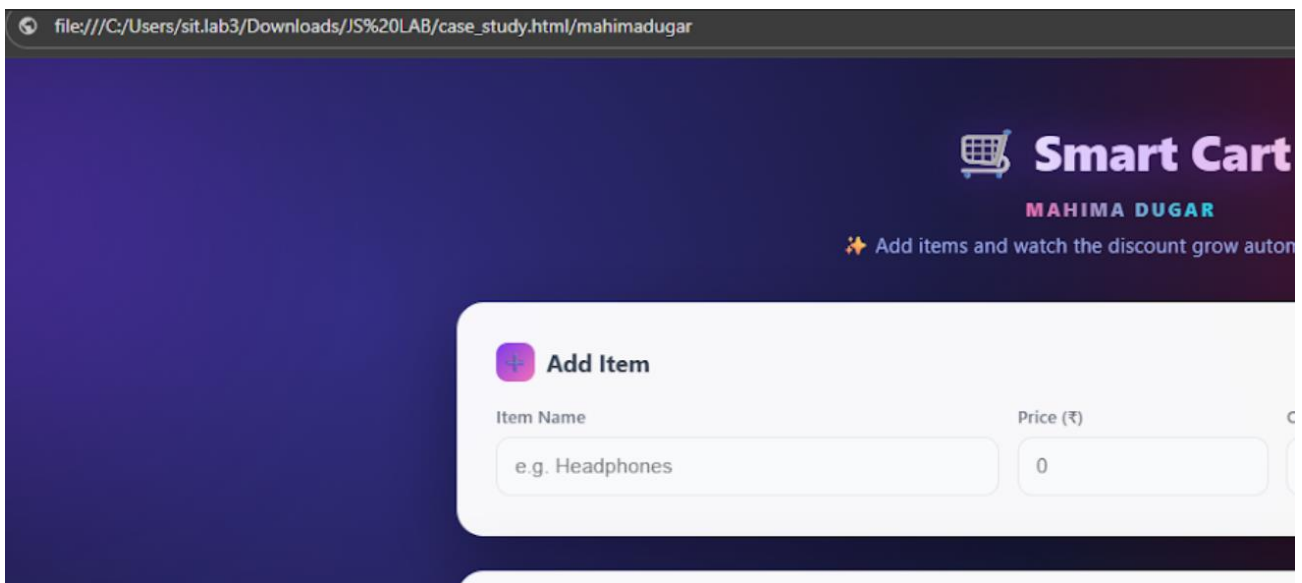
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Experiment No.: 05

- 1. Title:** Apply array methods and object handling; create a cart total calculator with discount logic.
- 2. Software/Tools Required:** HTML, VS Code.
- 3. Theory:** JavaScript arrays are used to store multiple values, while objects store data in **key-value pairs**. Array methods like `map ()`, `filter ()`, and `reduce ()` are useful for processing and calculating data. In a shopping cart, each product is represented as an object containing its name, price, and quantity. The `reduce ()` method is used to calculate the total cart value. Based on the total amount, discount conditions are applied using conditional statements, and the final payable amount is calculated.
- 4. Output:**





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Smart Cart

MAHIMA DUGAR

✦ Add items and watch the discount grow automatically ✦

Add Item

+ Add to Cart

Your Cart

ITEM	PRICE	QTY	SUBTOTAL	
headphones	₹1500.00	2	₹3000.00	X Remove

Bill Summary

TOTAL AMOUNT
₹3000.00

DISCOUNT (15%)
₹450.00

FINAL AMOUNT PAYABLE
₹2550.00

Discount Tiers • Above ₹5000: 20% | Above ₹2000: 15% | Above ₹1000: 10% | Above ₹500: 5% | Below ₹500: 0%

5.Result/Conclusion: The cart total calculator was successfully implemented using **JavaScript arrays, objects, array methods, and conditional discount logic**. The program calculates the total price, applies the appropriate discount, and displays the final amount to be paid.



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CASE STUDY

To study the functioning of Arrays and Loops in Java Script

- a) Create and manipulate arrays: push, pop, shift, unshift, splice, slice.
- b) Use array methods: map(), filter(), reduce(), forEach().

CaseStudy:-

Create an array of numbers and find the maximum and minimum value.

Code:

```
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4    <meta charset="UTF-8">
5    <meta name="viewport" content="width=device-width, initial-scale=1.0">
6
7    <title>JavaScript Arrays and Loops</title>
8
9    <style>
10     body {
11       font-family: Arial, sans-serif;
12       background-color: #f2f2f2;
13       margin: 0;
14       padding: 30px;
15     }
16
17     .container {
18       max-width: 900px;
19       margin: auto;
20       background: white;
21       padding: 30px;
22       border-radius: 12px;
23       box-shadow: 0 4px 15px rgba(0,0,0,0.15);
24     }
25
26     h1 {
27       text-align: center;
28       color: #333;
29     }
30
31     h2 {
32       color: #444;
33       border-bottom: 2px solid #ddd;
34       padding-bottom: 8px;
35     }
36
37     .section {
38       margin-bottom: 25px;
39       padding: 15px;
40       background: #fafafa;
41       border-radius: 8px;
42     }
```



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```
44  input {
45      padding: 10px;
46      width: 70%;
47      border: 1px solid #aaa;
48      border-radius: 5px;
49  }
50
51  button {
52      padding: 10px 15px;
53      margin: 5px;
54      border: none;
55      border-radius: 5px;
56      background-color: #333;
57      color: white;
58      cursor: pointer;
59  }
60
61  button:hover {
62      background-color: #555;
63  }
64
65  .output {
66      margin-top: 15px;
67      padding: 15px;
68      background-color: #e9e9e9;
69      border-radius: 5px;
70      line-height: 1.8;
71  }
72
73  .result {
74      font-weight: bold;
75      color: #222;
76  }
77  </style>
78  </head>
79
80  <body>
81
82  <div class="container">
83
84      <h1>JavaScript Arrays and Loops</h1>
85
86      <!-- Create Array -->
87      <div class="section">
88          <h2>Array Operations</h2>
89
90          <input type="text" id="numberInput"
91              | placeholder="Enter numbers separated by commas">
92
93          <br><br>
```

```
95      <button onclick="createArray()">Create Array</button>
96      <button onclick="pushElement()">Push</button>
97      <button onclick="popElement()">Pop</button>
98      <button onclick="shiftElement()">Shift</button>
99      <button onclick="unshiftElement()">Unshift</button>
100
101      <div class="output" id="arrayOutput">
102          | Array will appear here.
103      </div>
104  </div>
105
106  <!-- Splice and Slice -->
107  <div class="section">
108
109      <h2>Splice and Slice</h2>
110
111      <button onclick="spliceArray()">Splice</button>
112      <button onclick="sliceArray()">Slice</button>
113
114      <div class="output" id="spliceOutput">
115          | Result will appear here.
116      </div>
117  </div>
118
119
120  <!-- Array Methods -->
121  <div class="section">
122
123      <h2>Array Methods</h2>
124
125      <button onclick="mapArray()">map</button>
126      <button onclick="filterArray()">filter</button>
127      <button onclick="reduceArray()">reduce</button>
128      <button onclick="forEachArray()">forEach</button>
129
130      <div class="output" id="methodOutput">
131          | Result will appear here.
132      </div>
133  </div>
134
135
136  <!-- Case Study -->
137  <div class="section">
138
139      <h2>Case Study</h2>
140
141      <p>
142          | Create an array of numbers and find the maximum and minimum value.
143      </p>
```



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```
262
263
264 // REDUCE
265 function reduceArray() {
266
267     let result = numbers.reduce((sum, num) => sum + num, 0);
268
269     document.getElementById("methodOutput").innerHTML =
270         "<b>reduce() - Sum:</b> " +
271         result;
272 }
273
274 // FOREACH
275 function forEachArray() {
276
277     let result = "";
278
279     numbers.forEach(function(num) {
280         result += num + " ";
281     });
282
283     document.getElementById("methodOutput").innerHTML =
284         "<b>forEach() - Elements:</b> " +
285         result;
286 }
287
288 // CASE STUDY
289 function findMinMax() {
290
291     let maximum = Math.max(...numbers);
292     let minimum = Math.min(...numbers);
293
294     document.getElementById("minMaxOutput").innerHTML =
295         "<b>Array:</b> " + numbers.join(", ") +
296         "<br><b>Maximum:</b> " + maximum +
297         "<br><b>Minimum:</b> " + minimum;
298 }
299
300
301 // Display initial array
302 displayArray();
303
304
305 </script>
306
307 </body>
308 </html>
```

OUTPUT:

127.0.0.1:5500/jscasesstudy05.html/mahimadugar

JavaScript Arrays and Loops

Array Operations

12,23,45,56,78

Create Array Push Pop Shift Unshift

Current Array: 5, 23, 45, 56, 78

Splice and Slice

Splice Slice

After slice(1, 4): 23, 45, 56



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Array Methods

`map()` `filter()` `reduce()` `forEach()`

`forEach()` - Elements: 5 23 45 56 78

Case Study

Create an array of numbers and find the maximum and minimum value.

`Find Maximum & Minimum`

Array: 5, 23, 45, 56, 78

Maximum: 78

Minimum: 5

Result/Conclusion: The cart total calculator was successfully implemented using **JavaScript arrays, objects, array methods, and conditional discount logic**. The program calculates the total price, applies the appropriate discount, and displays the final amount to be paid.



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Experiment No.: 06

1. **Title:** Use String functions and regex for email validation, data extraction and text analysis.

2. **Software/Tools Required:** HTML, VS Code.

3. **Theory:** JavaScript string functions and Regular Expressions (Regex) are used for text processing, pattern matching, and data validation. String methods such as `length`, `toUpperCase()`, `toLowerCase()`, `substring()`, `replace()`, and `split()` help manipulate and analyze text data. Regex provides a powerful way to search, match, and validate patterns in strings.

In this practical, regex is used to validate email addresses by checking their format, while string functions are used for data extraction and text analysis, such as counting words, finding specific text, and replacing or extracting information from a string.

4. Output:

```
PS C:\Users\lenovo\Downloads\regex> & C:\Users\lenovo\AppData\Local\Programs\Python\Python314\python.exe c:/Users/lenovo/Downloads/regex/code.py
Valid Email

--- String Analysis ---
Length: 94
Uppercase: HELLO MAHIMA! I AM LEARNING PYTHON. MY MARKS ARE 85 AND 92. CONTACT ME AT MAHIMA123@GMAIL.COM.
Lowercase: hello mahima! i am learning python. my marks are 85 and 92. contact me at mahima123@gmail.com.
Word Count: 16
Python present: True
Python occurrences: 1
Numbers: ['85', '92', '123']
Emails: ['mahima123@gmail.com']
Modified Text: Hello Mahima! I am learning Programming. My marks are 85 and 92. Contact me at mahima123@gmail.com.
PS C:\Users\lenovo\Downloads\regex> |
```



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```
----- TEXT REPLACEMENT -----
Text after replacing Python with Programming:

Hello! My name is Mahima. I am a B.Tech CSE student.
I am learning Programming and Regular Expressions.
Programming is simple, powerful and useful for data analysis.
My marks are 85, 92 and 88.
You can contact me at mahima123@gmail.com.

----- SENTENCE ANALYSIS -----
Total Number of Sentences: 9
Sentence 1 : Hello
Sentence 2 : My name is Mahima
Sentence 3 : I am a B
Sentence 4 : Tech CSE student
Sentence 5 : I am learning Python and Regular Expressions
Sentence 6 : Python is simple, powerful and useful for data analysis
Sentence 7 : My marks are 85, 92 and 88
Sentence 8 : You can contact me at mahima123@gmail
Sentence 9 : com

----- MARKS EXTRACTION -----
Marks found: ['85', '92', '88']
Marks as numbers: [85, 92, 88]
Highest Marks: 92
```

5.Result: Successfully implemented email validation, data extraction, and text analysis using JavaScript string functions and Regular Expressions (Regex). The program correctly validates email formats, extracts required data, and performs text processing operations efficiently.



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CASE STUDY

AIM: To perform various String operations using Java Script.

a) To perform Basics of string.

b) Work with strings: substring(), indexOf(), split(), replace().

Case Study :-

1. Write a program to reverse a string.
2. WAP to count the number of vowels in a paragraph.

Code:

```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <meta name="viewport" content="width=device-width, initial-scale=1.0">
6
7   <title>JavaScript String Operations</title>
8
9   <style>
10     body {
11       font-family: Arial, sans-serif;
12       background-color: #f2f2f2;
13       margin: 0;
14       padding: 30px;
15     }
16
17     .container {
18       max-width: 900px;
19       margin: auto;
20       background-color: white;
21       padding: 30px;
22       border-radius: 12px;
23       box-shadow: 0 4px 15px rgba(0,0,0,0.15);
24     }
25
26     h1 {
27       text-align: center;
28       color: #333;
29     }
30
31     h2 {
32       color: #444;
33       border-bottom: 2px solid #ddd;
34       padding-bottom: 8px;
35     }
36
37     .section {
38       background-color: #fafafa;
39       padding: 20px;
40       margin-bottom: 20px;
41       border-radius: 8px;
42     }
43
```



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```
44     input, textarea {
45         width: 90%;
46         padding: 10px;
47         margin: 8px 0;
48         border: 1px solid #aaa;
49         border-radius: 5px;
50         font-size: 15px;
51     }
52
53     textarea {
54         height: 100px;
55         resize: vertical;
56     }
57
58     button {
59         padding: 10px 15px;
60         margin: 5px;
61         border: none;
62         border-radius: 5px;
63         background-color: #333;
64         color: white;
65         cursor: pointer;
66     }
67
68     button:hover {
69         background-color: #555;
70     }
71
72     .output {
73         margin-top: 15px;
74         padding: 15px;
75         background-color: #e9e9e9;
76         border-radius: 5px;
77         line-height: 1.8;
78     }
79 </style>
80 </head>
81
82 <body>
83
```

```
84 <div class="container">
85     <h1>JavaScript String Operations</h1>
86
87     <!-- Basic String Operations -->
88     <div class="section">
89
90         <h2>Basic String Operations</h2>
91
92         <input type="text" id="stringInput"
93             | placeholder="Enter a string">
94
95         <br>
96
97         <button onclick="stringLength()">Length</button>
98         <button onclick="toUpperCase()">Uppercase</button>
99         <button onclick="toLowerCase()">Lowercase</button>
100
101         <div class="output" id="basicOutput">
102             | Result will appear here.
103         </div>
104     </div>
105
106     <!-- String Methods -->
107     <div class="section">
108
109         <h2>String Methods</h2>
110
111         <button onclick="substringOperation()">substring()</button>
112         <button onclick="indexOfOperation()">indexOf()</button>
113         <button onclick="splitOperation()">split()</button>
114         <button onclick="replaceOperation()">replace()</button>
115
116         <div class="output" id="methodOutput">
117             | Result will appear here.
118         </div>
119     </div>
120
121 </div>
122
123 </body>
124
125 </html>
```



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```
126 <!-- Reverse String -->
127 <div class="section">
128
129     <h2>Case Study 1: Reverse a String</h2>
130
131     <input type="text" id="reverseInput"
132           placeholder="Enter a string">
133
134     <br>
135
136     <button onclick="reverseString()">Reverse String</button>
137
138     <div class="output" id="reverseOutput">
139         Result will appear here.
140     </div>
141
142 </div>
143
144
145 <!-- Count Vowels -->
146 <div class="section">
147
148     <h2>Case Study 2: Count Vowels</h2>
149
150     <textarea id="paragraphInput"
151             placeholder="Enter a paragraph"></textarea>
152
153     <br>
154
155     <button onclick="countVowels()">Count Vowels</button>
156
157     <div class="output" id="vowelOutput">
158         Result will appear here.
159     </div>
160
161 </div>
162
163 </div>
164
```

```
168 // BASIC STRING OPERATIONS
169
170 function stringLength() {
171
172     let str = document.getElementById("stringInput").value;
173
174     document.getElementById("basicOutput").innerHTML =
175         "<b>Length:</b> " + str.length;
176 }
177
178
179 function toUpper() {
180
181     let str = document.getElementById("stringInput").value;
182
183     document.getElementById("basicOutput").innerHTML =
184         "<b>Uppercase:</b> " + str.toUpperCase();
185 }
186
187
188 function toLower() {
189
190     let str = document.getElementById("stringInput").value;
191
192     document.getElementById("basicOutput").innerHTML =
193         "<b>Lowercase:</b> " + str.toLowerCase();
194 }
195
196
197 // SUBSTRING
198
199 function substringOperation() {
200
201     let str = document.getElementById("stringInput").value;
202
203     let result = str.substring(0, 5);
204
205     document.getElementById("methodOutput").innerHTML =
206         "<b>substring(0, 5):</b> " + result;
207 }
```



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```
268
269 // CASE STUDY 2
270 // COUNT VOWELS
271
272 function countVowels() {
273
274     let paragraph =
275         document.getElementById("paragraphInput").value;
276
277     let count = 0;
278
279     for (let char of paragraph.toLowerCase()) {
280
281         if ("aeiou".includes(char)) {
282             count++;
283         }
284     }
285
286     document.getElementById("vowelOutput").innerHTML =
287         "<b>Number of Vowels:</b> " + count;
288 }
289
290 </script>
291
292 </body>
293 </html>
```

Output:

127.0.0.1:5500/jscaestudy06.html/mahimadugar

JavaScript String Operations

Basic String Operations

hello , my name is mahima dugar ,currently in my 3rd year pursuing btech cse at sit nagpur.

Length Uppercase Lowercase

Length: 91

String Methods

substring() indexOf() split() replace()

split(): hello | , | my | name | is | mahima | dugar | ,currently | in | my | 3rd | year | pursuing | btech | cse | at | sit | nagpur.



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Case Study 1: Reverse a String

Reverse String

Original String: hellooooo

Reversed String: ooooooleh

Case Study 2: Count Vowels

Count Vowels

Number of Vowels: 24

Conclusion: Thus, we successfully performed various String operations in JavaScript such as `substring()`, `indexOf()`, `split()`, and `replace()`. We also implemented programs to reverse a string and count the number of vowels in a paragraph, gaining a better understanding of JavaScript string manipulation and loops.



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Experiment No.: 07

1. **Title:** Perform DOM traversal and updates; develop a to-do list app with add, edit, and delete features.

2. **Software/Tools Required:** HTML, VS Code.

3. **Theory:** DOM (Document Object Model) represents an HTML page as a tree of elements. JavaScript uses the DOM to **access, traverse, create, update, and delete** HTML elements dynamically.

In this experiment, DOM methods and event handling are used to develop a **To-Do List application**. The user can enter a task and add it to the list. Each task contains **Edit** and **Delete** options. The Edit option modifies the task, while Delete removes it from the DOM.

Key methods used: getElementById(), querySelector(), createElement(), appendChild(), textContent, and remove().

To-Do List Application

A to-do list is a simple application that demonstrates DOM manipulation and event handling. The user enters a task and clicks **Add** to create a new list item. Each task can have **Edit** and **Delete** buttons.

Working Logic:

1. Take the task entered by the user from the input field.
2. Create a new element using JavaScript.
3. Add the task text and Edit/Delete buttons to the list.
4. The **Edit** button allows the user to modify the existing task.
5. The **Delete** button removes the task from the DOM.
6. DOM methods are used to dynamically update the to-do list.



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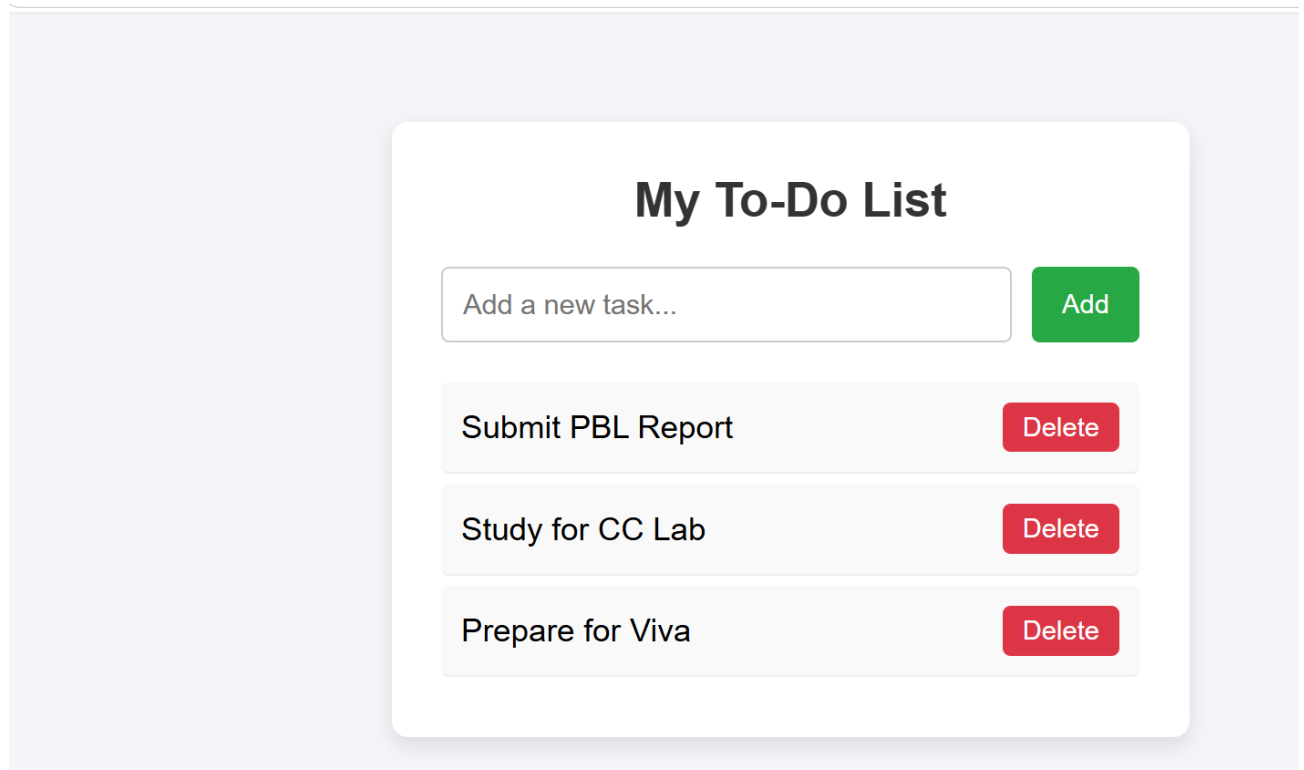
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4. Output:

127.0.0.1:5500/index.html/MAHIMADUGAR/24070521280



5.Conclusion: Thus, the experiment successfully demonstrates **DOM traversal and dynamic DOM updates using JavaScript**. A functional to-do list application was developed with **Add, Edit, and Delete** operations. This experiment helps in understanding how JavaScript interacts with HTML elements and dynamically changes webpage content based on user actions.



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CASE STUDY

AIM: To Demonstrate the Events Handling in JavaScript

- a) Access and manipulate form elements using JavaScript.
- b) Validate user input (e.g. email, empty fields).
- c) Use various form events (submit, change, focus).

Case Study:-

Create the following HTML form and use required JavaScript and events in a given form.

Code:

```
1 <!DOCTYPE html>
2 <html>
3 <head>
4   <title>JavaScript Event Handling</title>
5
6   <style>
7     body {
8       font-family: Arial, sans-serif;
9       background-color: #f2f2f2;
10      padding: 30px;
11    }
12
13    .container {
14      width: 500px;
15      margin: auto;
16      background-color: white;
17      padding: 25px;
18      border-radius: 10px;
19      box-shadow: 0 0 10px gray;
20    }
21
22    h1 {
23      text-align: center;
24    }
25
26    label {
27      display: block;
28      margin-top: 12px;
29    }
30
31    input, select {
32      width: 100%;
33      padding: 10px;
34      margin-top: 5px;
35      box-sizing: border-box;
36    }
37
38    button {
39      width: 100%;
40      padding: 12px;
41      margin-top: 20px;
42      background-color: #333;
43      color: white;
44      border: none;
45      cursor: pointer;
46    }
47
48    button:hover {
49      background-color: #555;
50    }
51
52    .error {
53      color: red;
54      font-size: 14px;
55    }
```



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```
63 </head>
64
65 <body>
66
67 <div class="container">
68
69 <h1>Registration Form</h1>
70
71 <form id="myForm">
72
73 <label>Name:</label>
74 <input type="text" id="name" placeholder="Enter your name">
75 <span class="error" id="nameError"></span>
76
77 <label>Email:</label>
78 <input type="text" id="email" placeholder="Enter your email">
79 <span class="error" id="emailError"></span>
80
81 <label>Age:</label>
82 <input type="number" id="age" placeholder="Enter your age">
83
84 <label>Course:</label>
85 <select id="course">
86 <option value="">Select Course</option>
87 <option value="CSE">Computer Science</option>
88 <option value="IT">Information Technology</option>
89 <option value="ECE">Electronics</option>
90 </select>
91
92 <button type="submit">Submit</button>
93
94 </form>
95
96 <div id="message"></div>
97
98 </div>
99
100
101 <script>
102
103 // Accessing form elements
104 let name = document.getElementById("name");
105 let email = document.getElementById("email");
106 let age = document.getElementById("age");
107 let course = document.getElementById("course");
108 let form = document.getElementById("myForm");
109
110
111 // FOCUS EVENT
112 name.addEventListener("focus", function() {
113   name.style.backgroundColor = "#e8f0fe";
114 });
115
```

```
2 <html>
65 <body>
01 <script>
28 form.addEventListener("submit", function(event) {
56   "Email is required";
57
58   valid = false;
59
60   } else if (!emailPattern.test(email.value)) {
61
62     document.getElementById("emailError").innerHTML =
63       "Enter a valid email";
64
65     valid = false;
66   }
67
68
69   // Age validation
70   if (age.value === "") {
71
72     alert("Please enter your age");
73     valid = false;
74   }
75
76
77   // Course validation
78   if (course.value === "") {
79
80     alert("Please select a course");
81     valid = false;
82   }
83
84
85   // Successful submission
86   if (valid) {
87
88     document.getElementById("message").innerHTML =
89       "Form submitted successfully!";
90
91     document.getElementById("message").style.color =
92       "green";
93
94     alert("Registration Successful!");
95
96     form.reset();
97   }
98
99   });
100
101 </script>
102
103 </body>
104 </html>
```



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Output:

127.0.0.1:5500/jscasestudy07.html



Registration Form

Name:

Email:

Age:

Course:

Select Course

Submit

Form submitted successfully!

Result: Thus, the experiment successfully demonstrates **DOM traversal and dynamic DOM updates using JavaScript**. A functional to-do list application was developed with **Add, Edit, and Delete** operations. This experiment helps in understanding how JavaScript interacts with HTML elements and dynamically changes webpage content based on user actions.



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Experiment No.: 08

1. Title: Access and validate form fields using events; design a gym admission form with live input checks.

2. Software/Tools Required: HTML, VS Code.

3. Theory: JavaScript is used to **access and validate form fields** and make web forms interactive. Form elements can be accessed using methods like `getElementById()` and `querySelector()`.

JavaScript **events** such as `input`, `blur`, `change`, and `submit` help detect user actions. Live validation checks the entered information immediately and displays an error message if the data is incorrect or incomplete.

In the Gym Admission Form, fields such as **Name, Age, Email, Phone Number, Gender, and Membership Plan** can be validated. JavaScript displays suitable error messages when the entered data is invalid and provides feedback immediately. This improves user experience and prevents incorrect or incomplete data from being submitted.

Key features:

- Access form elements using JavaScript DOM methods.
- Handle user actions using events.
- Perform live input validation.
- Display error messages dynamically.
- Prevent form submission when invalid data is entered.
- Ensure that required fields are properly filled.



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4. Output:

127.0.0.1:5500/index.html/mahimadugar_24070521280



Gym Admission

Join us and start your fitness journey!

Full Name *

✓ Looks good!

Email *

✓ Valid email!

Phone Number *

✓ Valid phone number!

Age *

✓ Valid age!

Membership Plan *

Yearly - ₹12000

✓ Plan selected.

Register for Gym

 **Registration successful! Welcome to the gym, Mahima Dugar!**

127.0.0.1:5500/index.html/mahimadugar_24070521280



Gym Admission

Join us and start your fitness journey!

Full Name *

✓ Looks good!

Email *

✓ Valid email!

Phone Number *

✓ Valid phone number!

Age *

Age must be between 16 and 80.

Membership Plan *

Yearly - ₹12000

✓ Plan selected.

Register for Gym

✗ Please fill all fields correctly.



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CASE STUDY

AIM: To design an **Online Job Application Form** using HTML, CSS, and JavaScript with **real-time form validation**, ensuring that the entered details are valid before submission.

Code:

```
1 <!DOCTYPE html>
2 <html>
3 <head>
4   <title>Online Job Application Form</title>
5
6   <style>
7     body {
8       font-family: Arial, sans-serif;
9       background-color: #f2f5f9;
10      margin: 0;
11      padding: 20px;
12    }
13
14    .container {
15      width: 75%;
16      margin: auto;
17      background: #fff;
18      padding: 25px;
19      border-radius: 10px;
20      box-shadow: 0 0 10px #gray;
21    }
22
23    h1 {
24      text-align: center;
25      color: #1f4e79;
26    }
27
28    h2 {
29      background-color: #1f4e79;
30      color: #fff;
31      padding: 8px;
32      font-size: 18px;
33    }
34
35    .form-group {
36      margin-bottom: 15px;
37    }
38
39    label {
40      display: inline-block;
41      width: 220px;
42      font-weight: bold;
43      vertical-align: top;
44    }
45
46    </style>
47
48    input, select, textarea {
49
50    }
51
52    textarea {
53      height: 70px;
54      resize: vertical;
55    }
56
57    .error {
58      color: red;
59      font-size: 13px;
60      margin-left: 225px;
61      display: block;
62      margin-top: 4px;
63    }
64
65    .success {
66      color: green;
67      text-align: center;
68      font-weight: bold;
69      margin-top: 15px;
70    }
71
72    .buttons {
73      text-align: center;
74      margin-top: 25px;
75    }
76
77    button {
78      padding: 10px 25px;
79      margin: 5px;
80      border: none;
81      border-radius: 5px;
82      cursor: pointer;
83      font-size: 15px;
84    }
85
86    #submitBtn {
87      background-color: #1f4e79;
88      color: #fff;
89    }
90
91  </html>
```



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```
104 <body>
106 <div class="container">
110 <form id="jobForm">
163 <!-- Educational Qualification -->
164 <h2>Educational Qualification</h2>
165
166 <div class="form-group">
167 <label>Highest Qualification:</label>
168 <select id="qualification">
169 <option value="">Select Qualification</option>
170 <option>Diploma</option>
171 <option>B.Tech / B.E.</option>
172 <option>B.Sc.</option>
173 <option>BCA</option>
174 <option>M.Tech / M.E.</option>
175 <option>MCA</option>
176 <option>M.Sc.</option>
177 <option>Other</option>
178 </select>
179 <span class="error" id="qualificationError"></span>
180 </div>
181
182 <div class="form-group">
183 <label>University / Institute:</label>
184 <input type="text" id="university" placeholder="Enter university/institute">
185 <span class="error" id="universityError"></span>
186 </div>
187
188 <div class="form-group">
189 <label>Passing Year:</label>
190 <input type="number" id="year" placeholder="YYYY">
191 <span class="error" id="yearError"></span>
192 </div>
193
194 <div class="form-group">
195 <label>Percentage / CGPA:</label>
196 <input type="text" id="cgpa" placeholder="Enter percentage or CGPA">
197 <span class="error" id="cgpaError"></span>
198 </div>
199
200
201 <!-- Professional Information -->
202 <h2>Professional Information</h2>
203
204 </div>
205 </body>
206 </html>
```

```
2 <html>
104 <body>
269 <script>
272 function showError(id, message) {
273   document.getElementById(id).textContent = message;
274 }
275
276 // Function to clear all errors
277 function clearErrors() {
278   let errors = document.querySelectorAll(".error");
279
280   errors.forEach(function(error) {
281     error.textContent = "";
282   });
283
284   document.getElementById("successMessage").textContent = "";
285 }
286
287
288 // Form submission event
289 document.getElementById("jobForm").addEventListener("submit", function(event) {
290
291   event.preventDefault();
292
293   clearErrors();
294
295   let valid = true;
296
297   // Get values
298   let name = document.getElementById("name").value.trim();
299   let dob = document.getElementById("dob").value;
300   let gender = document.getElementById("gender").value;
301   let email = document.getElementById("email").value.trim();
302   let mobile = document.getElementById("mobile").value.trim();
303   let address = document.getElementById("address").value.trim();
304   let location = document.getElementById("location").value.trim();
305
306   let qualification = document.getElementById("qualification").value;
307   let university = document.getElementById("university").value.trim();
308   let year = document.getElementById("year").value;
309   let cgpa = document.getElementById("cgpa").value.trim();
310
311   let position = document.getElementById("position").value.trim();
312   let experience = document.getElementById("experience").value.trim();
313   let organization = document.getElementById("organization").value.trim();
314 }
```



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```
104 <body>
269 <script>
289 document.getElementById("jobForm").addEventListener("submit", function(event) {
356     else if (!emailPattern.test(email)) {
357         showError("emailError", "Enter a valid email address.");
358         valid = false;
359     }
360
361
362     // Mobile validation
363     let mobilePattern = /^[0-9]{10}$/;
364
365     if (mobile === "") {
366         showError("mobileError", "Mobile number is required.");
367         valid = false;
368     }
369     else if (!mobilePattern.test(mobile)) {
370         showError("mobileError", "Enter a valid 10-digit mobile number.");
371         valid = false;
372     }
373
374
375     // Address validation
376     if (address === "") {
377         showError("addressError", "Address is required.");
378         valid = false;
379     }
380
381
382     if (location === "") {
383         showError("locationError", "City / State / PIN Code is required.");
384         valid = false;
385     }
386
387
388     // Education validation
389     if (qualification === "") {
390         showError("qualificationError", "Select highest qualification.");
391         valid = false;
392     }
393
394     if (university === "") {
395         showError("universityError", "University / Institute is required.");
```

```
2 </html>
104 <body>
269 <script>
289 document.getElementById("jobForm").addEventListener("submit", function(event) {
444     valid = false;
445 }
446
447     if (skills === "") {
448         showError("skillsError", "Technical skills are required.");
449         valid = false;
450     }
451
452     if (certifications === "") {
453         showError("certificationsError", "Certification details are required.");
454         valid = false;
455     }
456
457
458     // Final result
459     if (valid) {
460         document.getElementById("successMessage").textContent =
461             "Application submitted successfully!";
462         alert("Job Application Submitted Successfully!");
463     }
464 }
465
466 });
467
468
469 // Reset event
470 document.getElementById("jobForm").addEventListener("reset", function() {
471
472     setTimeout(function() {
473         clearErrors();
474     }, 10);
475
476 });
477
478 </script>
479
480 </body>
481 </html>
```



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Output:

127.0.0.1:5500/jscasestudy08.html

Online Job Application Form

Personal Information

Full Name:

Date of Birth:

Gender:

Email:

Mobile Number:

Address:

City / State / PIN Code:

Educational Qualification

Highest Qualification:

University / Institute:

Passing Year:

Percentage / CGPA:

Professional Information

Applying For / Job Position:

Total Work Experience:

Current / Previous Organization:

Current Designation:

Expected Salary:

Professional Information

Applying For / Job Position:

Total Work Experience:

Current / Previous Organization:

Current Designation:

Expected Salary:

Technical Skills

Programming Languages:

Technical Skills:

Certifications:



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Result: Thus, an **Online Job Application Form** was successfully developed using HTML, CSS, and JavaScript. JavaScript validation ensures that incorrect or incomplete information is detected before submission, making the form more reliable, user-friendly, and interactive.